

University of Information Technology & Sciences (UITs)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: BSc. in CSE
Mid Term Examination, Spring 2024
Course Title: Digital Logic Design
Course Code: CSE 215

Marks: 20

Time: 1 (One) Hour

(Answer all questions)

- 1 a) Explain De Morgan's Law with proper examples. In the following 2's complement addition/ subtraction, calculate the result and find in which scenario overflow occurred. [06]
- i) +5-3
 - ii) -8-5
 - iii) -3-4
 - iv) +5+6

- b) Interpret the multiplication of the following Hexadecimal numbers X and Y. [04]
- X=2A3C
Y=DB7

- 2 a) State the Absorption theorem of boolean algebra and prove this by algebraic manipulation with proper note. [02]
- b) Draw a logic diagram of EX-OR gate using NAND gate. [03]
- c) Analyze the following boolean function and express it in terms of canonical POS [05]

$$F(Q, R, S, T) = QS + R'T$$

A B C D

4 var k-map

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Time: 3(three)hours

(Answer all questions)

1. a) You have given a task with designing a traffic signal controller for a busy intersection. The controller has three inputs representing the traffic flow on each road: North-South (NS), East-West (EW), and Left Turn (LT). The traffic signal should allow the North-South traffic to flow unless there is heavy East-West traffic or a left turn request. In such cases, the signal should prioritize East-West traffic. [05]
 1. Devise the truth table and
 2. Simplify the above boolean function using K-map.
- b) Define Priority Encoder. Design 4:2 priority encoder with proper truth table and circuit diagram. [05]
2. a) Define BCD code, Excess-3 code and Reflected code. Design and draw the circuit diagram of BCD to Excess-3 converter with proper expression. [04]
- b) Sketch a full subtractor with half subtractor and inverter circuit with proper truth table, function simplification and design the circuit with only NOR gate. [06]
3. a) Explain 4 bit parallel adder and give two examples of combinational circuits that use this 4 bit adder with proper diagram. [06]
- b) Design a 4 bit equality comparator with circuit diagram. How a magnitude comparator finds the greater one from two values, suppose A=1001 and B=0111. [04]

4. a) You are designing a circuit for a security system in a high-tech facility. The security system is equipped with a keypad that allows users to enter a 4-digit passcode. To enhance security, you decide to implement a function that triggers an alarm if the entered passcode matches any of the following combinations: 0000, 0001, 0011, 0101, 1101, 1100, or 1101. [04]

Using a 2:4 decoder, **design** a circuit for the security system that activates the alarm when the correct passcode is entered. And also provide the truth table for your circuit design.

- b) **Demonstrate** the below boolean function using a 4:1 Mux. [03]
$$F(A,B,C,D) = \Sigma (2,4,6,7,9,14)$$

- c) **Discuss** the characteristics of combinational logic circuits and **demonstrate** the procedural steps of a combinational circuit for solving a specific problem. [03]

5. a) **Construct** the K-map of the following boolean function with the don't-care conditions d , express the simplified function in sum-of-minterms form and implement the circuit diagram with only NAND gate. [06]

$$F(A, B, C, D) = \Sigma m(1, 5, 6, 12, 13, 14) + d(2, 4)$$

- b) Define Sequential Circuit. **Draw** SR flip flop using gates and **explain** its operations. [04]